University of Central Punjab

(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)

FACULTY OF INFORMATION TECHNOLOGY

Introduction to Database System (Lab)

Graded Lab 2

Instructions

1. Create your own **database** using your Registration Number.
   - CREATE DATABASE **L1F20BSSEXXXX**;
   - USE **L1F20BSSEXXXX**;
2. Replace **XXXX** with your last 4 digits of your registration number.
3. Execute queries in sequence.
4. Use ALTER where required.
5. Submit **SQL queries and screenshots in the same word file**.

```sql
create database if not exists L1F24BSCS0506;
use L1F24BSCS0506;

-- Task - 01
CREATE TABLE Instructor (
    instructor_id VARCHAR(15),
    name VARCHAR(50),
    age INT,
    gender CHAR(1),
    office_adress VARCHAR(100),
    PRIMARY KEY (instructor_id)
);

-- Task - 02
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	08:22:51	set sql_safe_updates=0	0 row(s) affected	0.000 sec
2	08:22:53	create database x	1 row(s) affected	0.000 sec
3	08:28:16	create database if not exists L1F24BSCS0506	1 row(s) affected	0.000 sec
4	08:28:19	use L1F24BSCS0506	0 row(s) affected	0.015 sec

Q1. Instructor Table Creation

The university wants to store instructor details. Write a query to create a table **Instructor** with:

- InstructorID VARCHAR(15) (Primary Key)
- Name VARCHAR(50)
- Age INT
- Gender CHAR(1)
- OfficeAddress VARCHAR(100)

```sql
-- Task - 01
CREATE TABLE Instructor (
    instructor_id VARCHAR(15),
    name VARCHAR(50),
    age INT,
    gender CHAR(1),
    office_adress VARCHAR(100),
    PRIMARY KEY (instructor_id)
);

-- Task - 02
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	08:22:51	set sql_safe_updates=0	0 row(s) affected	0.000 sec
2	08:22:53	create database x	1 row(s) affected	0.000 sec
3	08:28:16	create database if not exists L1F24BSCS0506	1 row(s) affected	0.000 sec
4	08:28:19	use L1F24BSCS0506	0 row(s) affected	0.015 sec
5	08:58:32	CREATE TABLE Instructor (instructor_id VARCHAR(15), name VARCHAR(50), age INT, ...	0 row(s) affected	0.016 sec

Q2. Student Table Creation

Write a query to create a table **Student** with:

- StudentID VARCHAR(15) (Primary Key)
- Name VARCHAR(50)
- Age INT
- Gender CHAR(1)
- Address VARCHAR(100)
- Major VARCHAR(30)

---

l1f24bscs0506-graded-lab

Limit to 1000 rows

```
    age INT,
    gender CHAR(1),
    office_adress VARCHAR(100),
    PRIMARY KEY (instructor_id)
);

-- Task - 02
CREATE TABLE Student (
    student_id VARCHAR(15),
    name VARCHAR(50),
    age INT,
    gender CHAR(1),
    adress VARCHAR(100),
    major VARCHAR(30),
    PRIMARY KEY (student_id)
);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	08:22:53	create database x	1 row(s) affected	0.000 sec
3	08:28:16	create database if not exists L1F24BSCS0506	1 row(s) affected	0.000 sec
4	08:28:19	use L1F24BSCS0506	0 row(s) affected	0.015 sec
5	08:58:32	CREATE TABLE Instructor (instructor_id VARCHAR(15), name VARCHAR(50), age INT, ...	0 row(s) affected	0.016 sec
6	08:59:14	CREATE TABLE Student (student_id VARCHAR(15), name VARCHAR(50), age INT, ...	0 row(s) affected	0.016 sec

Q3. Add Column to Instructor Table

Add a column **Department VARCHAR(20)** to the Instructor table.

```
-- Task - 03
alter table Instructor
add column department varchar(20);

-- Task - 04
alter table Student
modify column major Varchar(50);

-- Task - 05
alter table Student
add column instructor_id varchar(15);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	08:28:16	create database if not exists L1F24BSCS0506	1 row(s) affected	0.000 sec
4	08:28:19	use L1F24BSCS0506	0 row(s) affected	0.015 sec
5	08:58:32	CREATE TABLE Instructor (instructor_id VARCHAR(15), name VARCHAR(50), age INT, ...	0 row(s) affected	0.016 sec
6	08:59:14	CREATE TABLE Student (student_id VARCHAR(15), name VARCHAR(50), age INT, ...	0 row(s) affected	0.016 sec
7	08:59:55	alter table Instructor add column department varchar(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec

Q4. Modify Column in Student Table

Change the datatype of **Major** from VARCHAR(30) to VARCHAR(50).

```
-- Task - 04
alter table Student
modify column major Varchar(50);

-- Task - 05
alter table Student
add column instructor_id varchar(15);

alter table Student
add constraint instructor_fk FOREIGN KEY(instructor_id) REFERENCES Instructor(instructor_id)
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	08:28:19	use L1F24BSCS0506	0 row(s) affected	0.015 sec
5	08:58:32	CREATE TABLE Instructor (instructor_id VARCHAR(15), name VARCHAR(50), age INT, ...	0 row(s) affected	0.016 sec
6	08:59:14	CREATE TABLE Student (student_id VARCHAR(15), name VARCHAR(50), age INT, ...	0 row(s) affected	0.016 sec
7	08:59:55	alter table Instructor add column department varchar(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec
8	09:00:36	alter table Student modify column major Varchar(50)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.000 sec

Q5. Establish Relationship Between Student and Instructor

Add **InstructorID VARCHAR(15)** in Student and make it a foreign key referencing Instructor.

```
-- Task - 05
alter table Student
add column instructor_id varchar(15);

alter table Student
add constraint instructor_fk FOREIGN KEY(instructor_id) REFERENCES Instructor(instructor_id)
on delete cascade on update cascade;

-- Task - 06
CREATE TABLE Registration_Record (
    reg_id VARCHAR(15),
    reg_date DATE,
```

Action Output

#	Time	Action	Message	Duration / Fetch
6	08:59:14	CREATE TABLE Student (student_id VARCHAR(15), name VARCHAR(50), age INT...	0 row(s) affected	0.016 sec
7	08:59:55	alter table Instructor add column department varchar(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec
8	09:00:36	alter table Student modify column major Varchar(50)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.000 sec
9	09:01:07	desc Student	6 row(s) returned	0.000 sec / 0.000 sec
10	09:02:15	alter table Student add column instructor_id varchar(15)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.000 sec
11	09:02:30	alter table Student add constraint instructor_fk FOREIGN KEY(instructor_id) REFERENCES I...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec

Q6. Course Registration Records

Create table **Registration**:

- RegID VARCHAR(15) (Primary Key)
- RegDate DATE
- Fee DOUBLE
- StudentID VARCHAR(15) (FK)
- InstructorID VARCHAR(15) (FK)

```sql
-- Task - 06
CREATE TABLE Registration_Record (
    reg_id VARCHAR(15),
    reg_date DATE,
    fee DOUBLE,
    student_id VARCHAR(15),
    instructor_id VARCHAR(15),
    PRIMARY KEY (reg_id),
    FOREIGN KEY (student_id) REFERENCES Student (student_id),
    FOREIGN KEY (instructor_id) REFERENCES Instructor (instructor_id)
);

-- Task - 07
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
15	09:06:32	CREATE TABLE Registration_Record (reg_id VARCHAR(15), reg_date DATE, fee ...	Error Code: 1005. Can't create table 11f24bscs0506`.`registration_record` (errno: 121 "Duplic...	0.015 sec
16	09:06:41	CREATE TABLE Registration_Record (reg_id VARCHAR(15), reg_date DATE, fee ...	Error Code: 1005. Can't create table 11f24bscs0506`.`registration_record` (errno: 121 "Duplic...	0.000 sec
17	09:07:41	CREATE TABLE Registration_Record (reg_id VARCHAR(15), reg_date DATE, fee ...	Error Code: 1005. Can't create table 11f24bscs0506`.`registration_record` (errno: 121 "Duplic...	0.000 sec
18	09:07:54	CREATE TABLE Registration_Record (reg_id VARCHAR(15), reg_date DATE, fee ...	Error Code: 1005. Can't create table 11f24bscs0506`.`registration_record` (errno: 121 "Duplic...	0.000 sec
19	09:08:12	CREATE TABLE RegistrationRecord (reg_id VARCHAR(15), reg_date DATE, fee D...	Error Code: 1005. Can't create table 11f24bscs0506`.`registrationrecord` (errno: 121 "Duplicat...	0.218 sec
20	09:09:39	CREATE TABLE Registration_Record (reg_id VARCHAR(15), reg_date DATE, fee ...	0 row(s) affected	0.015 sec

Q7. Rename Column in Registration

Rename **Fee** to **RegistrationFee**.

```
-- Task - 07
alter table Registration_Record
change column fee registration_fee double;

-- Task - 08
alter table Registration_Record
add column room_no varchar(10);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
16	09:06:41	CREATE TABLE Registration_Record (reg_id VARCHAR(15), reg_date DATE, fee ...	Error Code: 1005. Can't create table `11f24bscs0506`.`registration_record` (errno: 121 "Duplic...	0.000 sec
17	09:07:41	CREATE TABLE Registration_Record (reg_id VARCHAR(15), reg_date DATE, fee ...	Error Code: 1005. Can't create table `11f24bscs0506`.`registration_record` (errno: 121 "Duplic...	0.000 sec
18	09:07:54	CREATE TABLE Registration_Record (reg_id VARCHAR(15), reg_date DATE, fee ...	Error Code: 1005. Can't create table `11f24bscs0506`.`registration_record` (errno: 121 "Duplic...	0.000 sec
19	09:08:12	CREATE TABLE RegistrationRecord (reg_id VARCHAR(15), reg_date DATE, fee D...	Error Code: 1005. Can't create table `11f24bscs0506`.`registrationrecord` (errno: 121 "Duplicat...	0.218 sec
20	09:09:39	CREATE TABLE Registration_Record (reg_id VARCHAR(15), reg_date DATE, fee ...	0 row(s) affected	0.015 sec
21	09:10:28	alter table Registration_Record change column fee registration_fee double	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.000 sec

Q8. Add Column to Registration

Add **RoomNo VARCHAR(10)**.

```
-- Task - 08
alter table Registration_Record
add column room_no varchar(10);

-- Task - 09
CREATE TABLE Classroom (
    room_id VARCHAR(15),
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
18	09:07:54	CREATE TABLE Registration_Record (reg_id VARCHAR(15), reg_date DATE, fee ...	Error Code: 1005. Can't create table 11f24bscs0506`.`registration_record` (errno: 121 "Duplic...	0.000 sec
19	09:08:12	CREATE TABLE RegistrationRecord (reg_id VARCHAR(15), reg_date DATE, fee D...	Error Code: 1005. Can't create table 11f24bscs0506`.`registrationrecord` (errno: 121 "Duplicat...	0.218 sec
20	09:09:39	CREATE TABLE Registration_Record (reg_id VARCHAR(15), reg_date DATE, fee ...	0 row(s) affected	0.015 sec
21	09:10:28	alter table Registration_Record change column fee registration_fee double	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.000 sec
22	09:10:48	desc Registration_Record	5 row(s) returned	0.000 sec / 0.000 sec
23	09:11:26	alter table Registration_Record add column room_no varchar(10)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.094 sec

Q9. Classroom Allocation

Create table **Classroom**:

- RoomID VARCHAR(15) (Primary Key)
- RoomType VARCHAR(20)
- Capacity INT
- AllocationDate DATE

```sql
-- Task - 09
CREATE TABLE Classroom (
    room_id VARCHAR(15),
    room_type VARCHAR(20),
    capacity INT,
    allocation_date DATE,
    PRIMARY KEY (room_id)
);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
19	09:08:12	CREATE TABLE RegistrationRecord (reg_id VARCHAR(15), reg_date DATE, fee D...	Error Code: 1005. Can't create table `11f24bscs0506`.`registrationrecord` (errno: 121 "Duplicat...	0.218 sec
20	09:09:39	CREATE TABLE Registration_Record (reg_id VARCHAR(15), reg_date DATE, fee ...	0 row(s) affected	0.015 sec
21	09:10:28	alter table Registration_Record change column fee registration_fee double	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.000 sec
22	09:10:48	desc Registration_Record	5 row(s) returned	0.000 sec / 0.000 sec
23	09:11:26	alter table Registration_Record add column room_no varchar(10)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.094 sec
24	09:11:59	CREATE TABLE Classroom (room_id VARCHAR(15), room_type VARCHAR(20), ca...	0 row(s) affected	0.000 sec

Q10. Fee Records

Create table **StudentFee**:

- FeeID VARCHAR(15) (Primary Key)
- FeeDate DATE
- Amount VARCHAR(20)
- StudentID VARCHAR(15)

```sql
-- Task - 10
CREATE TABLE Student_Fee (
    fee_id VARCHAR(15),
    fee_date DATE,
    amount VARCHAR(20),
    student_id VARCHAR(15),
    PRIMARY KEY (fee_id)
);

-- Task - 11
alter table Student_Fee
ADD CONSTRAINT student_fk FOREIGN KEY(student_id) REFERENCES Student(student_id)
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
20	09:09:39	CREATE TABLE Registration_Record (reg_id VARCHAR(15), reg_date DATE, fee ...	0 row(s) affected	0.015 sec
21	09:10:28	alter table Registration_Record change column fee registration_fee double	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.000 sec
22	09:10:48	desc Registration_Record	5 row(s) returned	0.000 sec / 0.000 sec
23	09:11:26	alter table Registration_Record add column room_no varchar(10)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.094 sec
24	09:11:59	CREATE TABLE Classroom (room_id VARCHAR(15), room_type VARCHAR(20), ca...	0 row(s) affected	0.000 sec
25	09:12:39	CREATE TABLE Student_Fee (fee_id VARCHAR(15), fee_date DATE, amount VAR...	0 row(s) affected	0.000 sec

Q11. Establish Relationship

Make StudentID in StudentFee a foreign key.

```
-- Task - 11
alter table Student_Fee
ADD CONSTRAINT student_fk FOREIGN KEY(student_id) REFERENCES Student(student_id)
 ON DELETE CASCADE ON UPDATE CASCADE;

-- Task - 12
alter table Classroom
rename to Lecture_Room;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
21	09:10:28	alter table Registration_Record change column fee registration_fee double	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.000 sec
22	09:10:48	desc Registration_Record	5 row(s) returned	0.000 sec / 0.000 sec
23	09:11:26	alter table Registration_Record add column room_no varchar(10)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.094 sec
24	09:11:59	CREATE TABLE Classroom (room_id VARCHAR(15), room_type VARCHAR(20), ca...	0 row(s) affected	0.000 sec
25	09:12:39	CREATE TABLE Student_Fee (fee_id VARCHAR(15), fee_date DATE, amount VAR...	0 row(s) affected	0.000 sec
26	09:14:10	alter table Student_Fee ADD CONSTRAINT student_fk FOREIGN KEY(student_id) REFER...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec

Q12. Rename Table

Rename Classroom to **LectureRoom**.

```
-- Task - 12
alter table Classroom
rename to Lecture_Room;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
22	09:10:48	desc Registration_Record	5 row(s) returned	0.000 sec / 0.000 sec
23	09:11:26	alter table Registration_Record add column room_no varchar(10)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.094 sec
24	09:11:59	CREATE TABLE Classroom (room_id VARCHAR(15), room_type VARCHAR(20), ca...	0 row(s) affected	0.000 sec
25	09:12:39	CREATE TABLE Student_Fee (fee_id VARCHAR(15), fee_date DATE, amount VAR...	0 row(s) affected	0.000 sec
26	09:14:10	alter table Student_Fee ADD CONSTRAINT student_fk FOREIGN KEY(student_id) REFER...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
27	09:14:55	alter table Classroom rename to Lecture_Room	0 row(s) affected	0.000 sec

Q13. Modify Column

Change Amount to DOUBLE.

```
-- Task - 13
alter table Student_Fee
modify column amount double;

-- Task - 14
alter table Instructor
drop column office_adress;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
23	09:11:26	alter table Registration_Record add column room_no varchar(10)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.094 sec
24	09:11:59	CREATE TABLE Classroom (room_id VARCHAR(15), room_type VARCHAR(20), ca...	0 row(s) affected	0.000 sec
25	09:12:39	CREATE TABLE Student_Fee (fee_id VARCHAR(15), fee_date DATE, amount VAR...	0 row(s) affected	0.000 sec
26	09:14:10	alter table Student_Fee ADD CONSTRAINT student_fk FOREIGN KEY(student_id) REFER...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
27	09:14:55	alter table Classroom rename to Lecture_Room	0 row(s) affected	0.000 sec
28	09:15:32	alter table Student_Fee modify column amount double	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec

Q14. Drop Column

Drop OfficeAddress from Instructor.

```
-- Task - 14
alter table Instructor
drop column office_adress;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
24	09:11:59	CREATE TABLE Classroom (room_id VARCHAR(15), room_type VARCHAR(20), ca...	0 row(s) affected	0.000 sec
25	09:12:39	CREATE TABLE Student_Fee (fee_id VARCHAR(15), fee_date DATE, amount VAR...	0 row(s) affected	0.000 sec
26	09:14:10	alter table Student_Fee ADD CONSTRAINT student_fk FOREIGN KEY(student_id) REFER...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
27	09:14:55	alter table Classroom rename to Lecture_Room	0 row(s) affected	0.000 sec
28	09:15:32	alter table Student_Fee modify column amount double	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
29	09:16:08	alter table Instructor drop column office_adress	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec